

Experiment No. 3

Aim: Creating Transaction using Solidity and Remix IDE

Code:

```
// SPDX-License-Identifier: MIT

pragma solidity ^0.8.30;

contract Bank{

    // client account structure

    struct Client{

        address clientAddress;

        uint clientID;

        uint balance;

    }

    // manager address

    address payable public manager;

    // store client accounts

    mapping(address => Client) public clients;

    // client id counter

    uint nextClientID=1;
```

```
// set deployer as manager
```

```
constructor() {  
    manager=payable(msg.sender);  
}
```

```
// only manager can call
```

```
modifier onlyManager(){  
    require(msg.sender == manager, "Only manager can perform this action");  
    _;  
}
```

```
// Register a client
```

```
function registerClient() public {  
    require(clients[msg.sender].clientID == 0, "Client Already register");  
  
    clients[msg.sender]=Client(  
        msg.sender,  
        nextClientID,  
        0  
    );  
    nextClientID++;  
}
```

```
// deposit ether
```

```
function deposit() public payable {  
    require(clients[msg.sender].clientID !=0, "Register first");  
    require(msg.value>0,"Send some ether");  
    clients[msg.sender].balance += msg.value;  
}
```

```
// Withdraw ether - only manager
```

```
function withdraw(uint amount) public onlyManager {  
    require(amount <= address(this).balance, "Insufficient bank balance");  
  
    (bool success, ) = manager.call{value: amount}("");  
    require(success, "Transfer failed");  
}
```

```
// Get contract balance
```

```
function getBalance() public view returns (uint) {  
    return address(this).balance;  
}
```

```
// Receive ether directly
```

```
receive() external payable {
```

```
if (clients[msg.sender].clientID != 0) {  
    clients[msg.sender].balance += msg.value;  
}  
}  
}
```

Outputs:

Register client

Select a function to interact with... ▼

Low level interaction

+

● registerClient

Value

0

wei ▼

Gas limit

auto

0

Transact

Balance

0 ETH

✓

[vm] from: 0x5B3...eddc4 to: Bank.registerClient() 0xd91...39138 value: 0 wei data: 0x6d9...fb287 logs: 0 hash: 0x447...baca0

status

1 Transaction mined and execution completed

transaction hash

0x44743467fb1e9b588fe53f1c9ce9f4654cb52ae094964ea51bdbc97bdc3baca0

block hash

0xb7f7096054271bf8cc2ebe52a3e937d64d7b099117b9bef2170423044cf1a12b

block number

3

from

0x5B38Da6a701c568545dCfcB03FcB875f56beddC4

to

Bank.registerClient() 0xd9145CCE52D386f254917e481eB44e9943F39138

transaction cost

73341 gas

execution cost

52277 gas

output

0x

decoded input

{}

decoded output

{}

logs

[]

raw logs

[]

Select a function to interact with...

Low level interaction

● registerClient

Snipping Tool

Screenshot copied

Deposit amount:

Account 1 

98.999 ETH

0x5B3...eddc4 

0xd91...39138 

5min ago 

Functions

Select a function to interact with... 

Low level interaction

+

● deposit

Value 1 ether 

Gas limit auto 0

Transact

```
[vm] from: 0x5B3...eddC4 to: Bank.deposit() 0xd91...39138 value: 1000000000000000000 wei data: 0xd0e...30db0 logs: 0
hash: 0x8bb...28f96

status          1 Transaction mined and execution completed

transaction hash 0x8bbafelfb544f8e2625230d31045dbd7a5cc632366a846d08ce3b5bec5428f96 ⓘ

block hash      0xaf8190419fad4dac144bc640eaf49e2e001064f6ed59e1749bea7a2e0ac0166f ⓘ

block number    4 ⓘ

from            0x5B38Da6a701c568545dCfcB03FcB875f56beddC4 ⓘ

to              Bank.deposit() 0xd9145CCE52D386f254917e481eB44e9943F39138 ⓘ

transaction cost 45910 gas ⓘ

execution cost  24846 gas ⓘ

output          0x ⓘ

decoded input   {} ⓘ

decoded output  {} ⓘ

logs            [] ⓘ

raw logs        [] ⓘ

value           10000000000000000000 wei ⓘ
```

Get clients list:

clients

address

0x5B38Da6a701c568545dCfcB03FcB875f56beddC4

Calldata 

Parameters 

0:address: clientAddress 0x5B38Da6a701c568545dCfcB03FcB875f56beddC4

1:uint256: clientID 1

2:uint256: balance 100000000000000000000

Call

Get Balance:

Low level interaction

● **getBalance**

0:uint256: 10000000000000000000000

Call

Balance1.0 ETH

Withdraw amount:

● **withdraw** uint256

1

Calldata

Parameters

Value

0ether

Gas limit

auto0

Transact

Balance0.9999999999999999999999 ETH

```
✓ [vm] from: 0x5B3...eddC4 to: Bank.withdraw(uint256) 0xd91...39138 value: 0 wei data:
hash: 0x461...1b71d

status 1 Transaction mined and execution completed

transaction hash 0x46180510ccfbf679c8e0136d7c475abbd0a1981325b4dc56479ebe902bc1b71d ⓘ

block hash 0x875f2f207dbc478f1207b73293016b6fc3df4ac7d8aa43353952f4c3acc93bf0 ⓘ

block number 7 ⓘ

from 0x5B38Da6a701c568545dCfcB03FcB875f56beddC4 ⓘ

to Bank.withdraw(uint256) 0xd9145CCE52D386f254917e481eB44e9943F39138 ⓘ

transaction cost 31039 gas ⓘ

execution cost 9835 gas ⓘ

output 0x ⓘ

decoded input {
  "uint256 amount": "1"
} ⓘ

decoded output {} ⓘ

logs [] ⓘ

raw logs [] ⓘ
```

Checking balance again:

● getBalance

0:uint256: 999999999999999999999999

Call

Balance0.999999999999999999999999 ETH